Fiscal Year:	FY 2013	**Task Last Updated:**	FY 06/24/2013
PI Name:	Matson, Douglas Ph.D.		
Project Title:	Electromagnetic Levitation Flight Support for Transient Observation of Nucleation Events (ELFSTONE)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	douglas.matson@tufts.edu	**Fax:**	FY (617) 627-3058
PI Organization Type:	UNIVERSITY	**Phone:**	(617) 627-3239
Organization Name:	Tufts University		
PI Address 1:	Mechanical Engineering		
PI Address 2:	025 Anderson Hall, 200 College Ave		
PI Web Page:	https://		
City:	Medford	**State:**	MA
Zip Code:	02155	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	09/20/2010	**End Date:**	11/30/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	2
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	9	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 11/30/2013 per NSSC information (previously was changed to 9/19/2014, per NSSC information at that time) (Ed., 12/26/13) NOTE: New end date is 9/19/2014, per NSSC information (Ed., 6/24/13)		
Key Personnel Changes/Previous PI:	Douglas Matson is U.S. Team Member on ESA-sponsored projects, "Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC)" and "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)."		
COI Name (Institution):			
Grant/Contract No.:	NNX10AV27G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Funding is for Dr. Matson's role as Team Member on the subprojects described below, associated with ESA-sponsored main projects as noted. Main Project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany Subproject: Transformation kinetics - ISS Using results from thermophysical property evaluation and combining these with the MHD convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the MSL-EML to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany Subprojects: Management and Coordination. Coordination of the scientific program, Coordination of the ground based measurement program, Coordination of the microgravity experiment program, Contact with the national and international space agencies. Short Duration Microgravity Measurements - PF. Conduct parabolic flights on selected alloys. Long Duration microgravity Experiments - ISS. Surface tension and viscosity as a function of temperature, Density as a function of temperature, Solidification growth velocity. Preparation of the Microgravity Experiments. Experimental protocols for processing on the ISS , Verification of flight samples for the ISS. MODELLING A: Thermodynamics, Phase Selection. Thermodynamic phase diagram calculation, Modeling of the influence of convection on phase selection.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The value of this research is that the results have application to the design of industrial welding, spray forming, and casting operations for a commercially important class of soft magnetic materials. In addition, this research addresses fundamental issues relating to rapid solidification behavior, metastable phase selection, and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Task Progress:	International Working Group 15-16 November 2012 in Friedrichshafen DE. IWG 7 involved evaporation limit planning and payload status reporting. Visit to view flight hardware development by Astrium. Conducted flight support ESL viscosity tests at ELS-MSFC 7-18 January 2013. Conducted ground-based test sample evaluation of samples produced at the MSFC ESL in collaboration with MSFC M&P metallography staff. Conducted Parameter Set definition 28 Jan – 1 Feb 2013 by MUSC at DLR Köln DE for FeCo flight alloy samples 6 and 7. Conference organizer for TMS 2013 Materials Science in Reduced Gravity Symposium, San Antonio TX 6-8 March 2013. International Working Group 8-9 March 2013 in San Antonio TX. IWG 8 involved Batch 2 planning and software update recommendations. Graduated two MS students and two BS students. Completed Post-doc training for Jonghyun Lee and hired new Post-doc Vijaya Kumar. To conduct flight support ESL quenching tests at ESL-MSFC August 2013.
Bibliography Type:	Description: (Last Updated: 05/05/2026)
Articles in Peer-reviewed Journals	Matson DM. "Editorial notes: Materials science in reduced gravity." JOM Journal of the Minerals, Metals and Materials Society. 2012 Sep;64(9): 1087-8. , Sep-2012
Articles in Peer-reviewed Journals	Lee J, Matson DM. "Estimation of mass evaporation of Fe50Co50 during measurement of thermophysical properties using electrostatic levitation. " International Journal of Thermophysics. Accepted January 2013. In press as of June 2013. , Jun-2013
Awards	String G. "Tisch College Presidential Award for Citizenship and Public Service, May 2013." May-2013
Awards	Lee J. (Jonghyun Lee) "TMS EPD Young Leader Professional Development Award, March 2013." Mar-2013
Awards	String G. "2nd place Best Paper, 'Tufts Outreach Somerville,' ASEE NE section conference, Northfield VT, March 2013." Mar-2013

Awards	String G. "3rd place Best Poster, 'International Educationeers,' ASEE NE section conference, Northfield VT, March 2013." Mar-2013
Dissertations and Theses	Xiao X. (Xiao Xiao) "Magnetohydrodynamic Modeling of Convection inside Electromagnetically Levitated Molten Fe50Co50 Droplets." MS Thesis, Tufts University, May 2013. , May-2013
Dissertations and Theses	String G. (Gabrielle String) "An Evaluation of Student Implemented Clean Water Programs in Ecuador." MS Thesis, Tufts University, May 2013. , May-2013
Dissertations and Theses	Andrews T. (Tyler Andrews) "Design, construction and testing of an experimental apparatus for single optical pyrometer-based high-temperature differential thermal analysis for the study of rare earth material properties." BS Thesis, Tufts University, May 2013. , May-2013
Dissertations and Theses	Nakanishi B. (Brad Nakanishi) "Ultra-high temperature measurement and modeling of an induction-heated reactor for rapid molten oxide electrolysis of rare earths." BS Thesis, Tufts University, May 2013. , May-2013
Papers from Meeting Proceedings	String GM, Sander PJ, Maloney MM, Matson DM. "Tufts Outreach to Somerville High School's Engineering Classrooms: A Reflection of Year One." 2013 Northeast Conference of the American Society for Engineering Education (ASEE), Northfield, VT, March 14-16, 2013. 2013 Northeast Conference of the American Society for Engineering Education (ASEE), Northfield, VT, March 14-16, 2013. Paper 152. , Mar-2013